ENRIQUE GARCIA RIVERA

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EDUCATION

Washington University in St. Louis, McKelvey School of Engineering — *Expected: December 2026*
B.S. Mechanical Engineering — GPA: 3.64 / 4.00
Relevant Coursework: Modeling, Simulation & Control (MATLAB/Simulink), Numerical Methods & Matrix Algebra, Fluid Mechanics, Heat Transfer, Thermodynamics, Solid Mechanics, Vibrations, Electrical & Electronic Circuits

SKILLS

- **Software and Tools:** MATLAB/Simulink, Python (proficient), C++ (working knowledge), SolidWorks, ANSYS Mechanical, PX4 autopilot, Gazebo simulation, XFLR5, Git/version control, embedded systems (Teensy, Arduino)
- **Engineering Competencies:** Mechanical design and rapid prototyping, structural and aerodynamic analysis, sensor instrumentation and fusion, avionics systems integration, experimental testing and validation.

EXPERIENCE

WashU VTOL – *Avionics & Systems Integration Lead*, St. Louis, MO — Sep 2025 – Present
- Designed and integrated avionics architecture for a prototype eVTOL aircraft including Pixhawk 6C flight controller, GPS, IMU, telemetry radios, and power distribution systems.
- Configured PX4 autopilot and navigation sensors to support reliable communication between onboard systems and ground monitoring tools.
- Conducted staged subsystem validation through bench testing and HIL simulation, collecting and analyzing flight data to verify system stability and sensor performance.
- Diagnosed and resolved integration issues across sensors, computing hardware, and communication systems, documenting architecture and test procedures for repeatable troubleshooting.

MIRo Lab, UMKC – *Undergraduate Researcher (Grant Funded)*, Kansas City, MO — May – Aug 2025
- Built an embedded sensing platform synchronizing IMU and EMG sensor streams at 1 kHz on a Teensy 4.0, applying physical intuition about sensor behavior to design calibration and noise characterization procedures.
- Characterized sensor noise, bias drift, and timing jitter across motion capture and arm movement trials to evaluate measurement reliability.
- Collected and prepared IMU and EMG datasets across controlled motion trials to support Extended Kalman Filter development for multi-sensor state estimation.
- Developed Python scripts for signal processing, filtering, and statistical analysis to validate sensor fusion accuracy across experimental conditions.

WashU Design Build Fly – *Aerodynamics Engineer*, St. Louis, MO — Sep 2024 – Jan 2026
- Evaluated aircraft wing geometries using XFLR5 to analyze lift, drag, and stability tradeoffs, applying aerodynamic principles to support configuration selection.
- Performed structural finite element analysis in ANSYS Mechanical to verify wing spar load capacity and structural safety margins under expected flight loading.
- Designed and fabricated mechanical components including payload release mechanisms using SolidWorks CAD modeling and additive manufacturing.
- Supported flight test operations across multiple sessions, assisting with aircraft setup and logging flight data for post-flight review.

Washington University McKelvey School of Engineering – *Course Assistant*, St. Louis, MO — Aug – Dec 2025
- Supported MEMS 2000 Electrical & Electronic Circuits through laboratory sessions, instrumentation troubleshooting, and circuit debugging with students.

TECHNICAL PROJECTS

- **Hyperion Orbital Physics Simulator (C++):** Six-DOF orbital dynamics simulator implementing the J2 perturbation model — applied rigid body dynamics and numerical integration to study spacecraft motion and orbital decay.
- **Flight Navigation EKF System (Embedded C++):** Twelve-state Extended Kalman Filter combining GPS, barometer, and IMU measurements to estimate navigation states for autonomous systems in real time.
- **Sentinel Prototype (C++ and Python):** Designed and built a robotic sensing system integrating mechanical assembly, embedded closed-loop control, and multi-sensor feedback for experimental autonomous tracking.

INVOLVEMENT

Society of Hispanic Professional Engineers (SHPE) — Sep 2023 – Present
American Society of Mechanical Engineers (ASME) — Aug 2024 – Present
Kessler Scholars Collaborative, Washington University — Jun 2023 – Present